CS4084
Quantum Computing
Week 01 – Lecture 01

Dr. Maqsood M. Khan

Department of Computer Science
FAST-NUCES Peshawar

Outline

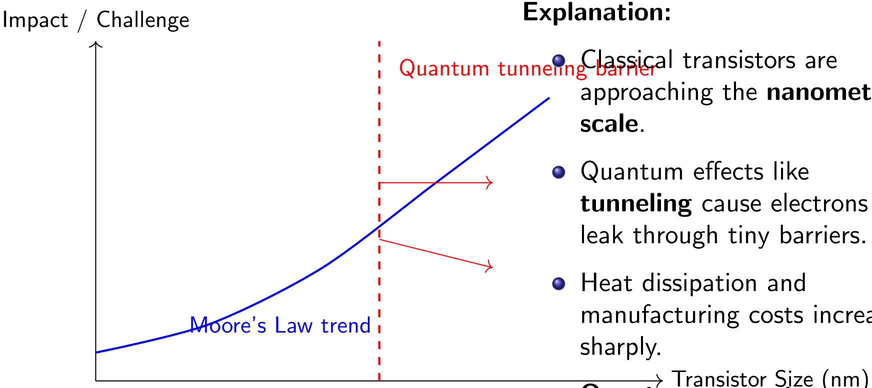
Introduction to Quantum Computing

- Quantum computing uses principles of **quantum mechanics** to process information.
- Motivated by limitations of classical computers in:
  - Scaling and transistor miniaturization
  - Power consumption
  - Computationally hard problems (e.g., factoring large numbers, simulating quantum systems)
- Quantum computers exploit:
  - **Superposition**
  - **Entanglement**
  - **Interference**

What is Quantum Mechanics?

- Quantum Mechanics (QM) studies the behavior of very small particles: electrons, photons, atoms.
- Classical physics does NOT apply at this scale.
- Key Principles:
  - **Superposition**: Particles can exist in multiple states simultaneously.
  - **Entanglement**: Particles can be linked; measuring one affects the other.
  - **Measurement**: Observing a quantum state collapses it to a definite outcome.
  - **Probability & Interference**: Quantum outcomes are probabilistic; interference can amplify correct results.
- **Relevance to quantum computing**: Qubits exploit these principles for faster and novel computation.

Limits of Classical Transistors: Motivation for Quantum Computing



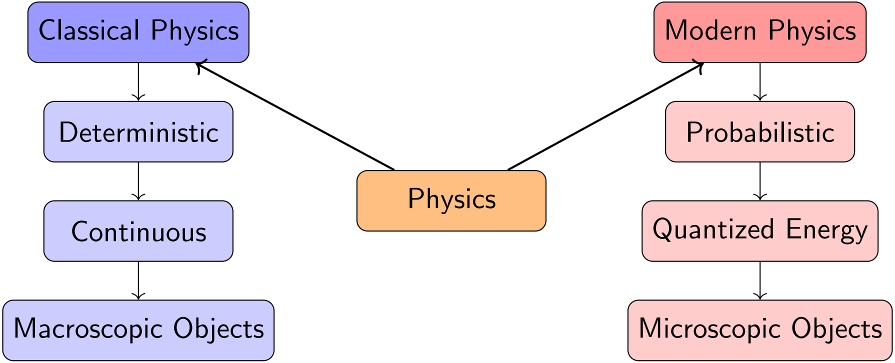
Explanation:

- Classical transistors are approaching the **nanometer scale**.
- Quantum effects like **tunneling** cause electrons to leak through tiny barriers.
- Heat dissipation and manufacturing costs increase sharply.
- **Quantum computers bypass these limits** by using qubits, which exploit superposition and entanglement.

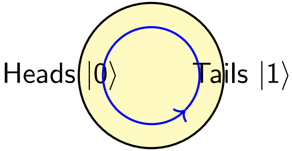
Classical vs Modern Physics - Key Differences

Feature	Classical Physics	Modern Physics
Determinism	Predictable	Probabilistic
State of Particle	Definite (position, velocity)	Superposition of states
Energy	Continuous	Quantized
Scale	Macroscopic	Microscopic / Relativistic
Key Idea	Newton's Laws	Quantum Mechanics / Relativity
Analogy	Light switch (on/off)	Spinning coin in the air

Classical vs Modern Physics - Key Concepts



Qubits: Spinning Coin Analogy



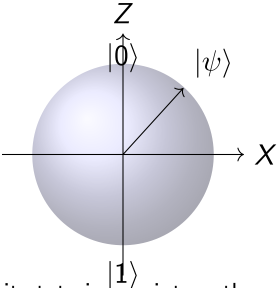
Coin in the air = Superposition

- Measurement collapses coin $\rightarrow$ shows Heads **or** Tails.
- A qubit can exist in **both states simultaneously**:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- $|\alpha|^2$ = probability of measuring $|0\rangle$, $|\beta|^2$ = probability of $|1\rangle$

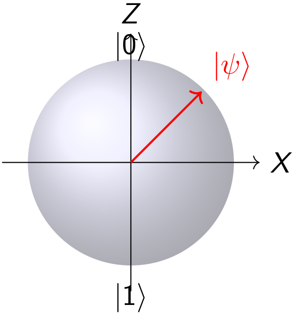
Qubits: Bloch Sphere Representation



Any qubit state is a point on the sphere

- Rotation on Bloch sphere = quantum gate operation.
- Geometric representation helps visualize superposition and phase.

Bloch Sphere: What Does It Tell Us?



- Bloch sphere represents the **state of a single qubit**.
- Each point on the sphere corresponds to:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- Top $= |0\rangle$, Bottom $= |1\rangle$
- Any other point $=$ **superposition**
- Quantum gates $=$ **rotations of the state vector**

Understanding Superposition: Qubits in Multiple States

- **Classical bit:** Like a light switch — can be either **0 (OFF)** or **1 (ON)**.
- **Qubit (Quantum bit):** Can exist in **both 0 and 1 simultaneously** — this is called **superposition**.

Analogy: Spinning Coin

- A classical coin lying flat = heads OR tails → like classical bit.
- A spinning coin in the air = heads AND tails at the same time → like a qubit in superposition.
- Catching the coin (measurement) = forces the coin to show either heads OR tails → collapses the superposition into a definite state.

Key Point: Qubits in superposition allow quantum computers to process **many possibilities simultaneously**.

Physical Implementations of Qubits

Qubit Type	**Physical System**	**Qubit States**	**Notes / Examples**
Superconducting Qubits	Tiny superconducting circuits	Different energy levels	IBM, Google, Rigetti
Trapped Ion Qubits	Single ions trapped with EM fields	Electronic energy levels	High precision, long coherence
Photonic Qubits	Single photons	Polarization, phase, or path	Quantum communication
Spin Qubits	Electron or nuclear spin	Spin up/down ($\|0\rangle$ / $\|1\rangle$)	Quantum dots, NV centers in diamond
Topological Qubits	Anyons in special materials	Braiding states	Robust against decoherence (research stage)

History and Milestones

- 1981 – Richard Feynman: simulating physics using quantum systems
- 1985 – David Deutsch: Quantum Turing Machine
- 1994 – Peter Shor: Factoring algorithm (threat to RSA)
- 1996 – Lov Grover: Search algorithm (quadratic speed-up)
- 2010s–present: NISQ devices by IBM, Google, IonQ

Classical vs Quantum Computation

Feature	Classical	Quantum
Unit of Info	Bit (0 or 1)	Qubit ($\lvert 0\rangle$, $\lvert 1\rangle$, superposition)
State Representation	Deterministic	Superposition of multiple states
Computation	Step-by-step	Parallel via superposition
Operations	Boolean gates	Quantum gates (Hadamard, CNOT, etc.)
Applications	General-purpose	Cryptography, simulation, optimization

Qubits and Quantum States

- Qubit: basic unit of quantum information
- State: $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$
- $\alpha, \beta \in \mathbb{C}$, $|\alpha|^2 + |\beta|^2 = 1$
- Measurement collapses qubit:
  - $|0\rangle$ with probability $|\alpha|^2$
  - $|1\rangle$ with probability $|\beta|^2$
- Multiple qubits can be **entangled**, sharing correlations

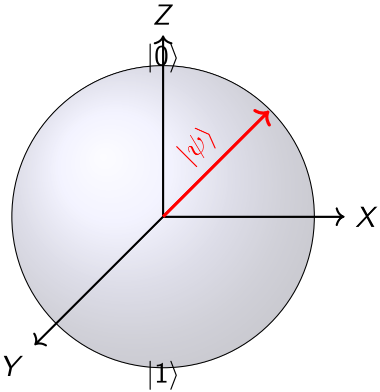
Bloch Sphere Representation

- Geometric representation of a single qubit:

$$|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle$$

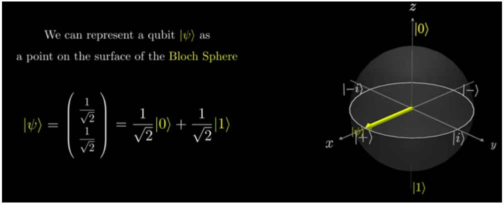
- Z-axis: $|0\rangle$ (north), $|1\rangle$ (south)
- X-axis / Y-axis: superposition states
- Visualizes rotations (quantum gates) and qubit states

Bloch Sphere Representation



Qubit state: $|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle$ Red arrow shows the qubit vector on the Bloch Sphere.

Bloch Sphere: Qubit Example



Qubit: $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

Probabilities:

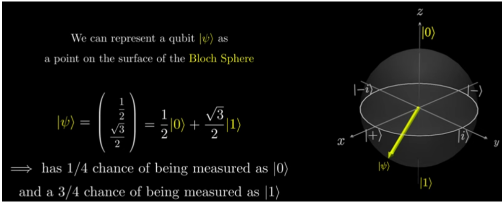
- $P(0) = |\alpha|^2 = 0.5$
- $P(1) = |\beta|^2 = 0.5$

Measurement:

- Qubit collapses to $|0\rangle$ or $|1\rangle$
- Equal chance (50

Arrow lies on equator $\rightarrow$ perfect superposition.

Bloch Sphere: Qubit Example (Unequal Superposition)



Qubit: $|\psi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle$

Probabilities:

- $P(0) = |\alpha|^2 = \left(\frac{1}{2}\right)^2 = 0.25$
- $P(1) = |\beta|^2 = \left(\frac{\sqrt{3}}{2}\right)^2 =$ 0.75

Measurement:

- Qubit collapses to $|0\rangle$ (25
- Arrow points closer to $|1\rangle$ on the Bloch sphere

Unequal superposition $\rightarrow$ higher probability for —1$\rangle$.

From Qubit to Qiskit

Concept	What It Means
Qubit	Basic unit of quantum information
$\|0\rangle, \|1\rangle$	Basis states of a qubit
Superposition	Qubit exists in multiple states at once
Quantum Gate	Operation that changes qubit state
Bloch Sphere	Visual model of a qubit state
Quantum Circuit	Sequence of quantum gates
Qiskit	Software used to build and run quantum circuits

- Qiskit lets us convert **quantum theory into executable circuits**.
- The same circuit can be:
  - Simulated on a classical computer
  - Executed on a real quantum computer

Key Concepts

- **Superposition**: Qubits in multiple states simultaneously
- **Entanglement**: Correlated qubits, even at distance
- **Interference**: Probability amplitudes reinforce or cancel
- **Measurement**: Collapses qubit to classical state
- **Quantum gates**: Manipulate qubits reversibly

Applications and Emerging Trends

- **Cryptography**: Shor's Algorithm, Quantum Key Distribution
- **Optimization**: Quantum annealing, QUBO problems
- **Machine Learning**: Speed-up in data analysis
- **Simulation**: Chemistry, physics, materials
- Research trends:
  - Error mitigation and correction
  - Scalable NISQ devices
  - Hybrid classical-quantum algorithms

From Assembly Language to Quantum Computing

Computer Organization	**Quantum Computing**
8086 Processor	Quantum Processing Unit (QPU)
Registers (AX, BX, IP)	Qubits
Bits (0 / 1)	Superposition States
Assembly Instructions	Quantum Gates
NASM Assembler	Qiskit Framework
DOSBox Emulator	Quantum Simulator
Physical CPU	Cloud Quantum Hardware

- DOSBox helped us understand low-level execution.
- Qiskit plays the same role for quantum computers.
- Difference: measuring a qubit changes its value.

Will Quantum Computers Replace Classical Computers?

Short Answer:

No — Quantum computers will *not* replace classical computers.

Why?

- Quantum computers are **special-purpose machines**, not general-purpose.
- They excel at specific problems:
  - Factoring large numbers (cryptography)
  - Quantum system simulation (chemistry, materials)
  - Optimization and search problems
- Classical computers are better for:
  - Operating systems
  - Programming, browsing, gaming
  - Everyday computing tasks

Key Idea: Quantum computers will act as **accelerators**, working alongside classical computers.

Analogy: CPU and GPU

- CPU = General-purpose processor
- GPU = Specialized accelerator
- GPU improves performance but does NOT replace CPU

Quantum Computing Analogy

- Classical Computer = CPU
- Quantum Computer = Accelerator (like GPU)

Key Idea: Quantum computers enhance specific tasks only.

Which Computer for Which Problem?

Problem Type	**Best Fit**
Web browsing, OS tasks	Classical
Databases, Networking	Classical
Cryptography (factoring)	Quantum
Molecular simulation	Quantum

Message: Different tools for different problems.

Summary

- Quantum computing leverages quantum mechanics to process information
- Different from classical computing in **representation, computation, and capabilities**
- Core concepts: qubits, superposition, entanglement, Bloch Sphere
- Current Focus: NISQ devices, algorithms, and research projects